NEC µPD6383GF — unofficial datasheet

> ⚠ **THIS IS NOT AN NEC DOCUMENT.** It is a reverse-engineering result, assembled from ROM corpora, service-manual schematics and live emulator traces by the Technics preservation project. NEC published no datasheet for this part that anyone has found — 0 hits for `638x` across bitsavers' complete NEC data-book holdings — and the only vendor documentation located is a block diagram and pin table in the Pioneer CDJ-500 service manual, where the part is IC302. Every claim carries a grade: **MEASURED**, **PROVEN BY CONSTRUCTION**, **INFERRED**, or **OPEN**. Nothing here is a guess dressed as a fact, and where something is unknown the entry says so rather than omitting the row. **Source of truth:** `dsp/instruction-set.md` in the disassembly repository, which is kept in step with the Python disassembler and with MAME's `upd6383d.cpp`. This page can drift from it; that one is authoritative.

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1. Overview

A fixed-point audio effects DSP, part of NEC's µPD638x consumer-audio line. Its immediate predecessor by part number, the µPD6382GF, is listed by distributors as a "*19-Bit Digital Signal Processor, QFP-80*"; no databook covers either.

Package	**100-pin QFP**. Pin 1 bottom-left, numbering counter-clockwise — MEASURED
Clock	**25 MHz** (KN5000, X301) — MEASURED
Frame rate	**44.1 kHz**, hardware-restarted — MEASURED four ways
Cycles per frame	25 MHz / 44.1 kHz = **567**, against 384 words of instruction RAM
Instruction word	**36 bits**, in a 5-byte big-endian right-aligned container; bits 36-39 always zero — MEASURED
Data width	**24-bit**; coefficients signed **Q0.23** — MEASURED
External memory	**17 address lines, 16 data lines** for delay DRAM — MEASURED from the pinout
Known part suffixes	`µPD6383GF-3BA`; a dealer lists it against Panasonic house number `GGC1163`

Known applications. Technics SX-KN5000 (IC311), SX-KN1500 (IC3), SX-WSA1 / SX-WSA1R (IC5, IC6, IC30 — three per machine), Pioneer CDJ-500 / CDJ-500G (IC302).

The chip is used **in pairs of effect "units"** in the Technics machines: one microprogram holds a common header, a call to unit 0's body, a call to unit 1's body, and a shared output stage.

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2. Pinout

Read at 400 dpi from the SX-WSA1R service manual, sheet II-15/II-16, where IC30 is drawn complete. The full table with wiring notes is in `dsp/hardware/upd6383-pinout.md`.

Pins 1-30 — bottom edge: host port, reset, serial audio

pin	name	pin	name	pin	name
1	`/CS`	11	`/BR-RQ`	21	`DI2`
2	`/C/D`	12	`/BR-AK`	22	`DI3`
3	`/SCK`	13	`/Fs-RST`	23	`DO1`
4	`SI`	14	`/Fs-MASK`	24	`DO2`
5	`SO`	15	`VDD`	25	`DO3`
6	`EIFLAG`	16	`GND`	26	`BCLKO`
7	`EOFLAG`	17	`BCLKI` ⚠	27	`LRCKO`
8	`RDY`	18	`LRCKI`	28	`XFsO1`
9	`/RST`	19	`XFsI`	29	`XFsO2`
10	`/RST2`	20	`DI1`	30	`TEST`

⚠ Pin 17's number is obscured in the drawing by a ground stub; `BCLKI` is placed there because the run 15/16/[17]/18 is otherwise unbroken at uniform pitch. Every other cell was read directly.

Pins 31-50 — right edge: mode straps, oscillator, DRAM control

pin	name	pin	name	pin	name	pin	name
31	`/EROF`	36	`GND`	41	`GND`	46	`A0`
32	`MD1`	37	`EOSC`	42	`VDD`	47	`A1`
33	`MD2`	38	`SEL`	43	`/RAS`	48	`A2`
34	`MD3`	39	`XI`	44	`/CAS`	49	`A3`
35	`MD4`	40	`XO`	45	`/WE`	50	`A4`

Pins 51-80 — top edge: DRAM address and data

pins	name
51-62	`A5` … `A16`, ascending with pin number
63	`VDD`
64	`GND`
65-80	`I/O1` … `I/O16`, ascending with pin number

Pins 81-100 — left edge: flags and parallel host port

pin	name	pin	name
81	`GND`	89	`/P/S`
82	`VDD`	90	`/WRITE`
83	`GF1`	91	`/READ`
84	`GF2`	92-99	`D0` … `D7`
85	`GF3`	100	`SETRDY`
86	`RQ1`		
87	`RQ2`		
88	`RQ3`		

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3. Functional description

3.1 Host interface — two of them, selected by `/P/S`

The chip offers a **serial** and a **parallel** host port, and the two known Technics designs use different ones — which is how we know the pin is a selector.

	serial	parallel
pins	`/CS` 1, `/C/D` 2, `/SCK` 3, `SI` 4, `SO` 5, `RDY` 8	`D0-D7` 92-99, `/WRITE` 90, `/READ` 91
used by	**SX-KN5000** (IC311)	**SX-WSA1R** (IC30, with `/P/S` tied to +5 V)

`/C/D` selects command versus data. `RDY` is open-drain — on the KN5000 it is pulled up by a 4.7 kΩ to +5 V and leaves the sheet as the net `DSPRDY`.

⚠ **The host port is write-oriented.** No route has been found by which the host reads a DSP result back, which is why every measurement of the chip's behaviour so far has had to come out through the audio path.

3.2 Flags — `RQ1-3` in, `GF1-3` out

From the CDJ-500 pin table: `RQ1`–`RQ3` are **host-written and testable by an instruction's condition field**, and `GF1`–`GF3` are **set by instructions and host-readable**.

⛔ **Neither the condition field nor the flag-setting instruction has been located in any microcode**, and neither Technics board makes them usable:

board	`RQ1-3`	`GF1-3`
SX-KN5000	strapped	—
SX-WSA1R (IC30)	**all three tied together to GND**	**unconnected stubs**

So on the machines we can reach, the host can neither vary `RQ` nor read `GF`. ⇒ predication is **OPEN and unobservable here**, and the correct statement is "*not available on this*

board".

3.3 Clock, reset and frame timing

`XI` / `XO` (39/40) are the oscillator pins and `EOSC` (37) is an external-oscillator input. `/RST` (9) and `/RST2` (10) are two resets. **`/Fs-RST` (13) restarts the program counter every sample** — there is no software frame loop, and the per-frame instruction budget is fixed by the clock ratio. `/Fs-MASK` (14) gates that. `XFsI` (19) is the frame-sync input and `XFsO1` / `XFsO2` (28/29) the outputs, so several chips can be chained on one frame clock.

`MD1` – `MD4` (32-35) are mode straps. **What they select is OPEN** — no source documents it. On the WSA1R's IC30, `MD1` is grounded and `MD2` / `MD3` / `MD4` go to +5 V.

`SETRDY` (100) and `/BR-RQ` (11) gate an emulator mode. On the KN5000 `SETRDY` is left open and `BR-RQ` is strapped high, which disables it: "*no PC trace without board modification*". What the mode delivers is **OPEN**; that it exists is the reason the pins are named as they are. `/BR-AK` (12) is the matching acknowledge.

3.4 Audio interface

`BCLKI` (17) / `LRCKI` (18) clock in; `BCLKO` (26) / `LRCKO` (27) clock out. **Three serial data inputs `DI1` - `DI3` (20-22) and three outputs `DO1` - `DO3` (23-25)**, so one chip can take three stereo lanes from a tone generator and return three processed lanes.

3.5 Memories

space	size	addressing	grade
Instruction RAM	384 words	host-loaded; PC restarts each frame	MEASURED
Coefficient RAM	256 cells, signed Q0.23	an implicit **cursor**, +1 per coefficient-consuming word, reset by one instruction	MEASURED; the whole cell map is recovered
Data RAM	per-unit state	an 8-bit **wrapping data pointer** with a signed post-increment from the `addr8` field	MEASURED; origin pinned 85/85
Register file	indexed by `addr8`, bit 7 = unit	named cells include per-unit output level and state-block base	MEASURED
External delay DRAM	17 address lines	`address = (descriptor[cursor] + G) mod 2^N`	PROVEN BY CONSTRUCTION

The delay arena is split per unit — on the KN5000, 32 768 words each = **743.0 ms** per unit. A delay line's length is the **difference of two descriptor cells**, and the ROM ships `15435 = 350 × 44100/1000` exactly, which is how the arithmetic was confirmed.

⚠ `N` in `mod 2^N` is **OPEN**: the firmware never uses more than 16 bits, so it is unobservable from software. The chip provides 17 address pins, which bounds it from the silicon side.

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4. Instruction word

```
 35             24 23  20 19        12 11                     0
+-----------------+------+------------+------------------------+
|      hi12       |class4|   addr8    |          lo12          |
+-----------------+------+------------+------------------------+
```

4.1 `hi12` is a horizontal microword, not an opcode — MEASURED

Its 54 observed values contain **77 Hamming-distance-1 pairs** against a popcount-matched null of 43.4 ± 4.3 (**z = +7.9**), spread over all twelve bit positions.

bit	meaning	grade
11	**FORMAT ESCAPE** — bits [10:0] mean something else	MEASURED
10	**END OF BLOCK** (when bit 11 clear); the word still does its datapath work	MEASURED
9:8	`f98` — a proven **field**, meaning **UNKNOWN**; arity 3	field MEASURED, meaning OPEN
7	gates bit 4's store; its own meaning **OPEN**	12 minimal pairs
6	third bit of the alternate-encoding flag	MEASURED, 0 exceptions in 6441 words
5	no reading; rendered as residue	OPEN
4	**STORE** accumulator → `mem[ptr]`, **before** the word's own ALU step, gated by bit 7	MEASURED; timing FORCED
3:1	`f31` — **the accumulator operation**	see below
0	"`addr8` is an absolute immediate"	PROVEN BY CONSTRUCTION for one family

`f31` (`hi12[3:1]`) — 8 of 8 values observed, **three anchored**:

value	operation	grade
0	`acc ← P` (LOAD)	anchored
1	`acc += P` (ADD)	anchored
2	`acc` unchanged (HOLD)	anchored
3, 4, 5, 6, 7	**OPEN**	189 words blocked on them

4.2 `class4` — addressing mode and cursor fetch — MEASURED

Bit 23 enables a **coefficient-cursor fetch** (fetch is not advance — only class `0xA` advances the cursor). Bits 2:0 are the addressing mode, and `hi12` **bit 11 picks the space**:

mode	meaning
0	pointer, no move
1	**register file** (without escape) / **external delay DRAM** (with escape, 324/324)
2	pointer with **post-increment by** `s8(addr8)` — the only mode that moves the pointer, and never escaped: 0 of 2399
3-6	uncharacterised — OPEN

4.3 `lo12` — operand routing

```
 11 10           6 5 4              0
+--+--------------+-+----------------+
|G |     SRC      |M|     ACTION     |
+--+--------------+-+----------------+
```

Bit 11 selects a second encoding with no SRC and no ACTION field — the device executes nothing for such a word. 169 undecoded words carry it; this is the single largest open block.

SRC — 7 of 18 observed codes anchored: `0x07` `mem[ptr]`, `0x10` accumulator, `0x19` tempA, `0x1A` tempB, `0x08` coefficient, plus `0x00` and `0x0B` which are **split by the word's own class**. `0x11` is the second accumulator `accb` (located, but its two write forms are not yet separable). `0x1C` is the effect's control bus — LFO for modulation programs, envelope for dynamics; it is **not LFO-specific**.

ACTION — 9 of 24 observed codes anchored: `0x00` (accumulator input term), `0x07` (`mem[ptr] ← bus`), `0x12` (no side effect), `0x13` (tempA), `0x14` (tempB), `0x15`, `0x19`, `0x0D` / `0x0E`. `ACT 0x0B` is anchored **only on class A**, as an all-pass multiplicand route.

⚠ How `ACT 0x12` and `ACT 0x15` differ is **OPEN**; both read as "no side effect".

4.4 The ALU — VERIFIED

```
L := src[ lo12[10:6] ]
if hi12 bit 4 and not (bit 7 and f31 != 2):
    mem[p] <- acc ; acc := 0                store AND clear, BEFORE the operation
acc := SRC_TERM + P_TERM                    ONE ADDER, TWO SELECTORS
         SRC_TERM = L    if lo12[4:0] == 0
                  = 0    if f31 == 0
                  = acc  otherwise
         P_TERM   = 0    if f31 == 2
                  = P    otherwise
if class4 == A:      P := (coef[cursor++] * L) >> 6
if class4 & 7 == 2:  p += (s8)addr8
```

The single adder is **FORCED**: two independent passes reached opposite orderings, and both demand the expression `bus + P`, which no ordering delivers and one adder does.

Fixed point — MEASURED live: `P = (coef[N-1] × L[N]) >> 6` with the coefficient latched one word early and the operand one slot late (`L[N] = mem[N-1]`); one-slot accumulator pipeline; store `datum = acc >> 16`.

⚠ The **total** scale is contested — shift 22 rails 19 of 48 biquad state cells where shift 23 rails none, while a counter-proposal is refuted on other evidence. The *relative* datapath is not in doubt.

4.5 Control flow — PROVEN BY CONSTRUCTION

Hardware PC restart per frame; a resident kernel of two canned blobs (header + output stage); a **two-level call stack** with entry addresses in host-loaded registers, not a vector table; and **straight-line, hand-unrolled bodies** — an exhaustive field scan for a branch instruction is negative, and one program repeats 32 words at period 8 varying only `addr8`.

⚠ Loop counters `LC1`-`LC3` appear in the pin table's feature list but no encoding has been found. One word = one slot, 285 slots per frame, holds in all 91 programs.

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5. What is not known

unknown	why it resists
`f31` 3, 4, 5, 6, 7	Separable and reproducible, but their carrier programs are ones no anchored criterion can see
The `lo12` bit-11 alternate encoding	169 words; addressing and accumulator function already characterised, the encoding itself is the one unknown
`f98` (`hi12[9:8]`)	Proven a field; the accumulator-op-selector reading was tested and failed
`hi12` bits 0, 5, 7	Bit 7 provably gates the store, but has no meaning of its own yet
`ACT 0x0B` off class A	Three readings survive; three separate criteria measured blind
`SRC 0x11`'s two write forms	A structural dependency cycle — every reachable program feeds it a frame-invariant constant
Class 6's table index	The word that computes it is a bit-11 word; six index sources enumerated and refuted
Addressing modes 3-6	No characterised twin
`COND` / predication	Unobservable on both Technics boards (§3.2)
`MD1`-`MD4` mode straps	Undocumented anywhere
DRAM read latency	Forced to a window; the exact value needs the emulator mode
`N` in `mod 2^N`	Unobservable from software; 17 address pins bound it

Coverage today: 80.8 % of 7273 distinct pooled microwords. The route from here is in state of play and the route to 100 %.

Related

Page	Description
State of play, and the route to 100 %	What is left and how to get it
Effects DSP (NEC uPD6383GF)	The full technical reference
SX-WSA1 / SX-WSA1R	The machine the pinout was read from